

Final Exam

Spring 2026

Total Time (min): 3 hr

Total Marks: 50

To

Total Questions: 3

Applied Human Computer Interaction (CS 3014)

Date: 5th June, 2026

Course Instructor(s)

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Roll No

Section

Student Signature

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NOTE: Attempt all the questions. Do the questions in order, out of order questions will not be marked

There are five questions in this paper

Do not use generic explanations, use clear and relevant HCI concepts in all answers

CLO 2: Apply user-centered design principles to design interactive interfaces

Q1a) Differentiate between **toolkits** and **User Interface Management Systems (UIMS)** on the basis of flexibility and level of systems they support (high/low) only? **[5 marks]**

Q1b) A bank develops an advanced mobile banking application for pensioners that has been thoroughly tested by the engineering team. The application performs all functions correctly, including money transfer, balance inquiry, and bill payment. No software defects are found during testing.

Which discipline, Software Engineering or Human-Computer Interaction, is better suited to address application's usability and why? **[5 marks]**

CLO 3: Analyze usability issues and evaluate interfaces using standard HCI techniques

Q2a) A team is evaluating two versions of an online learning platform — one with a traditional menu and one with a conversational chatbot interface. Explain why a *between-subjects design* would be more appropriate than a within-subjects design for this evaluation. In your answer, describe the experimental procedure and identify which variables must be controlled. **[5 marks]**

Q2b) A software company has developed a new mobile application for online ticket booking. Before involving real users, the development team wants to evaluate the interface.

The team has the following concerns:

- Whether first-time users can easily understand how to book a ticket. [2+1+1+1 marks]
- Whether the interface follows established usability principles.
- Whether experts can compare the design with existing usability guidelines and literature.
- Whether user performance can be predicted mathematically without conducting user tests.

Based on the classification of Analytic Evaluation Techniques, identify the most appropriate technique for each concern and justify your answer.

CLO 3: Analyze usability issues and evaluate interfaces using standard HCI techniques

Q3) A software company in 1960 developed a computer system for city planners. The system produces large paper reports containing maps, traffic statistics, and infrastructure data. City planners complain that it takes too much time to understand the information and make decisions.

In 1963, a new engineer proposes replacing paper reports with a display-screen system inspired by Ivan Sutherland's Sketchpad [5+5 (marks)]

Questions:

- a. What contribution of Sketchpad was established regarding the Video Display Unit only?
- b. What is the second important idea that was generated by this invention of Sutherland?

CLO 3: Analyze usability issues and evaluate interfaces using standard HCI techniques

Q4) A university is developing a **Student Information System (SIS)** to manage student records. The system maintains information such as student profiles, course registrations, grades, attendance, and fee status.

The university requires the system to provide:

1. A Student Portal where students can view their grades and attendance..
2. A Faculty Portal where instructors can enter grades and attendance records.
3. An Administrative Dashboard where staff can manage student records and generate reports.

Whenever a faculty member updates a student's grade, the change should immediately be reflected in all relevant interfaces. The university also plans to introduce additional dashboards in the future without changing the core student database. [5+5 (marks)]

Questions:

- a. How the Student Information System can be implemented using the MVC and PAC architectures. Clearly represent all components graphically.
- b. Which implementation model, MVC or PAC, is more suitable for this system? Justify your answer based on the requirements given in the scenario. .

CLO 3: Analyze usability issues and evaluate interfaces using standard HCI techniques

Q5) “Equipped with a dictionary, you are perfectly able to check your spelling by reading through the entire paper and correcting mistakes as you spot them. This mundane task is perfectly suited to automation, as the computer can check words against its own list of acceptable spellings”. [5+5 (marks)]

Questions:

- a. Considering Spell-checking application, to which usability principle and sub principle this scenario indicates?
 - b. According to the identified principle in part a, classify whether the application is Human operated or Computer operated with justification.
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THE END